

Curves: A curve can be parametrized by a function $y = f(x)$ for $a \leq x \leq b$.

Arc-Length Formula: If f' is continuous on $[a, b]$, then the length L of the curve $y = f(x)$ is given by

$$L = \int_a^b \sqrt{1 + (f'(x))^2} dx.$$

Example: Find the length of the curve given by

$$f(x) = \frac{2}{3} (1+x^2)^{3/2}$$

from $x = 0$ to $x = 2$.

$$L = \int_0^2 \sqrt{1 + (f'(x))^2} dx$$

$$L = \int_0^2 \sqrt{1 + 4x^2(1+x^2)} dx$$

$$= \int_0^2 \sqrt{4x^4 + 4x^2 + 1} dx$$

$$= \int_0^2 \sqrt{(2x^2 + 1)^2} dx$$

$$= \int_0^2 (2x^2 + 1) dx = \left. \frac{2x^3}{3} + x \right|_0^2 = \left(\frac{16}{3} + 2 \right) - 0 = \frac{22}{3}$$

constant multiple

$$f'(x) = \frac{2}{3} \cdot \frac{3}{2} (1+x^2)^{1/2} \cdot 2x$$

(chain rule)

Now we need to find $(f'(x))^2$

$$f'(x) = (1+x^2)^{1/2} \cdot 2x \rightarrow (f'(x))^2 = 4x^2(1+x^2)$$

Example: Find the length of the curve given by

$$f(x) = \ln(\cos(x))$$

from $x = 0$ to $x = \frac{\pi}{4}$.

$$L = \int_0^{\pi/4} \sqrt{1 + (f'(x))^2} dx$$

$$L = \int_0^{\pi/4} \sqrt{1 + \tan^2(x)} dx = \int_0^{\pi/4} \sqrt{\sec^2(x)} dx$$

$$= \int_0^{\pi/4} \sec(x) dx$$

$$= \ln|\sec(x) + \tan(x)| + C \Big|_0^{\pi/4}$$

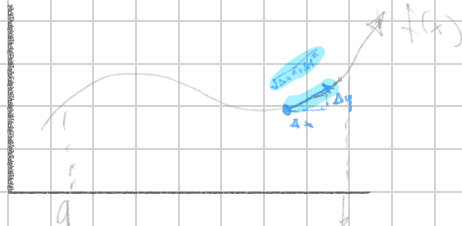
$$= \ln|\sec(\frac{\pi}{4}) + \tan(\frac{\pi}{4})| - \ln|\sec(0) + \tan(0)|$$

$$\int \sec(x) dx = \ln|\sec(x) + \tan(x)| + C$$

$$f'(x) = \frac{1}{\cos(x)} \cdot (-\sin(x))$$

$$= \frac{-\sin(x)}{\cos(x)} = -\tan(x)$$

$$(f'(x))^2 = \tan^2(x)$$



A curve is a function
 $y = f(x)$ over $a \leq x \leq b$

Arc Length

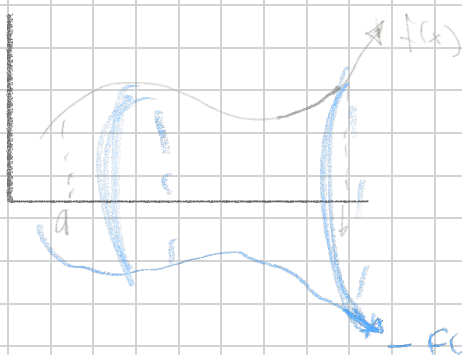
$$\sqrt{\Delta x^2 + \Delta y^2} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2} \cdot \Delta x$$

Take $\Delta x \rightarrow 0$ (so we are looking at vanishingly small pieces)

$$\text{Arc-length} = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_a^b \sqrt{1 + (f'(x))^2} dx$$

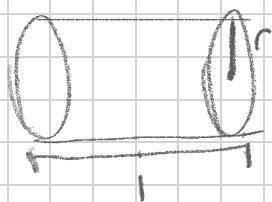
Using Pythagorean theorem infinitely many times over the interval $[a, b]$

Surfaces of Revolution



Building 3D
 shapes by
 revolving a curve
 around its axis

If $f(x) = r$, surface are as



$$S = 2\pi r l$$

S takes this form

$$2\pi \int_a^b \underbrace{f(x)}_{\substack{\text{radius} \\ = r(x)}} \underbrace{\sqrt{1 + (f'(x))^2}}_{l = \text{arc length}} dx$$

Surface of Revolution: A *surface of revolution* can be constructed by rotating a curve $y = f(x)$ about an axis.

Surface Area Formula: If f is positive and f' is continuous on $[a, b]$, then the surface area S of the surface of revolution generated by the curve $y = f(x)$ is given by

$$S = 2\pi \int_a^b f(x) \sqrt{1 + (f'(x))^2} dx.$$

Example: Find the surface area when $y = \sqrt{x}$, from $x = 0$ to $x = 1$ is rotated about the x -axis.

$r = \sqrt{x}$

$$2\pi \int_0^1 \sqrt{x} \sqrt{1 + (f'(x))^2} dx$$

$$= 2\pi \int_0^1 \sqrt{x} \sqrt{1 + \frac{1}{4x}} dx$$

$$= 2\pi \int_0^1 \sqrt{x + \frac{1}{4}} dx$$

$u = x + \frac{1}{4} \quad du = dx$

$$= 2\pi \int u^{\frac{1}{2}} du = 2\pi \frac{u^{\frac{3}{2}}}{\frac{3}{2}} = \frac{4\pi}{3} \left(x + \frac{1}{4} \right)^{\frac{3}{2}} \Big|_0^1$$

$$= \frac{4\pi}{3} \left(\left(\frac{5}{4} \right)^{\frac{3}{2}} - \left(\frac{1}{4} \right)^{\frac{3}{2}} \right)$$

Ignore bounds for now

Example: Find the surface area when $x = 1 + 2y^2$, from $y = 1$ to $y = 2$ is rotated about the x -axis.

$$1 + 2(1)^2 = 3 \rightarrow \text{when } y = 1, x = 3$$

$$1 + 2(2)^2 = 9 \rightarrow \text{when } y = 2, x = 9$$

We also need to insert
function in terms of x

$$x = 1 + 2y^2$$

$$\rightarrow y = \sqrt{\frac{x-1}{2}}$$

$$f(x) = \sqrt{\frac{x-1}{2}}$$

So we need bounds in terms of x

$$f'(x) = \frac{1}{2} \left(\frac{x-1}{2} \right)^{-\frac{1}{2}} \cdot \frac{1}{2}$$

$$S = 2\pi \int_3^9 \sqrt{\frac{x-1}{2}} \sqrt{1 + (f'(x))^2} dx$$

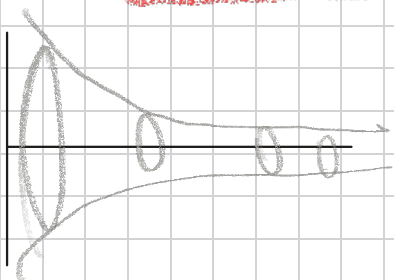
$\uparrow$
 $f(x) = \sqrt{\frac{x-1}{2}}$

Exercise: *Gabriel's Horn* is a surface of revolution obtained by rotating the curve $f(x) = \frac{1}{x}$ for $x \geq 1$ about the x -axis. Show that this shape has finite volume but infinite surface area.

Hint: Use the Comparison Test for infinite integrals.

See next page

Exercise: Gabriel's Horn is a surface of revolution obtained by rotating the curve $f(x) = \frac{1}{x}$ for $x \geq 1$ about the x -axis. Show that this shape has finite volume but infinite surface area. Hint: Use the Comparison Test for infinite integrals.



$$V = \pi \int_1^{\infty} f(x)^2 dx = \pi \int_1^{\infty} \frac{1}{x^2} dx$$

$$= \pi \lim_{C \rightarrow \infty} \int_1^C \frac{1}{x^2} dx = \pi \lim_{C \rightarrow \infty} \left. -\frac{1}{x} \right|_1^C = 0 - \pi(-1) = \pi$$

Finite

$$S = 2\pi \int_1^{\infty} \frac{1}{x} \sqrt{1 + (f'(x))^2} dx$$

$$= 2\pi \lim_{C \rightarrow \infty} \int_1^C \frac{1}{x} \sqrt{1 + (f'(x))^2} dx$$

Use comparison test